Maple Code Documentation for the Key Vanishing Lemma

Compact Case

March 27, 2026

Introduction

This document provides a detailed, line-by-line explanation of the Maple code used in the paper *Improved Degree Bounds for Hyperbolicity of Surfaces and Curve Complements* by Lei Hou, Dinh Tuan Huynh, Joël Merker, and Song-Yan Xie (arXiv:2603.14881). Specifically, this code implements the computation for the compact case of the Key Vanishing Lemma. The code systematically verifies the non-existence of nonzero negatively twisted invariant 2-jet differentials of order $m = 7$ on a degree 17 hypersurface in $\mathbb{P}^3$ (the code variable $\mathtt{d}$ is set to 16, so the actual degree is $\mathtt{d}+1 = 17$), subject to a specified twist parameter $\mathsf{Twist} = -7$. The computation involves solving algebraic systems to enforce the vanishing of the differential to high order along a hyperplane section. The code presented here is the full working version; the reader can copy and execute each block sequentially in a Maple worksheet to reproduce the results.

Part 1: Initialization and Parameter Definition

Explanation: Set the basic parameters. The command `restart` clears the Maple session. The variable $m = 7$ is the order of the jet differential. The twist parameter is set to -7. The variable `d` is set to 16; this means the actual degree of the hypersurface is $\mathtt{d} + 1 = 17$. The variable `degre` is defined as `d`+1 but is not directly used in the computation.

```
restart:
m := 7;
twist := -7;
d := 16;
degre := d + 1:
```

Listing 1: Initialization code

Part 2: Defining the Auxiliary Polynomial

Explanation: Define the auxiliary polynomial $TT = x^{\mathtt{d}+1} + 1^{\mathtt{d}+1} + 1 \cdot x^{\lfloor \mathtt{d}/2 \rfloor}$. Since $\mathtt{d} = 16$, we have $\mathtt{d} + 1 = 17$ and $\lfloor \mathtt{d}/2 \rfloor = 8$. Thus $TT = x^{17} + 1 + x^8$. Later, the full hypersurface equation will be $RR = x^{17} + y^{17} + z^{17} + TT = x^{17} + y^{17} + z^{17} + x^8 + 1$.

```
TT := x^(d+1) + 1^(d+1) + 1*x^floor(d/2);
```

Listing 2: Auxiliary polynomial definition

Part 3: Template for the Jet Differential

Explanation: Construct the local expression of an invariant jet differential J_{yx} on the affine chart $\{R_z \neq 0\}$. It consists of sums over k (the power of the Wronskian Δ_{yx}) and j (the power of y_1). The coefficients AA_j, BB_j, CC_j, DD_j are unknown polynomials in y (and later x, z) that will be determined. The factors $1/Rz^{m-k\cdot 2}$ ensure the appropriate homogeneity. Here $m = 7$, and we compute:

- $k = 0$: $m - 0 \cdot 2 = 7$, $m - 0 \cdot 3 = 7$
- $k = 1$: $m - 1 \cdot 2 = 5$, $m - 1 \cdot 3 = 4$
- $k = 2$: $m - 2 \cdot 2 = 3$, $m - 2 \cdot 3 = 1$
- $k = 3$: $m - 3 \cdot 2 = 1$, $m - 3 \cdot 3 = -2$

The sums range over $j = 0$ to $m - 3k$. Thus:

- For $k = 0$: j from 0 to 7
- For $k = 1$: j from 0 to 4
- For $k = 2$: j from 0 to 1
- For $k = 3$: j from 0 to -2 (empty sum, so no terms)

```
Jetyx := expand(
    + (1/Rz^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (y1)^j * Delta_yx^0, j=0..m-0*3)
    + (1/Rz^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (y1)^j * Delta_yx^1, j=0..m-1*3)
    + (1/Rz^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (y1)^j * Delta_yx^2, j=0..m-2*3)
    + (1/Rz^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (y1)^j * Delta_yx^3, j=0..m-3*3)
);
```

Listing 3: Jet expression construction

Part 4: Degree Bounds for Unknown Polynomials

Explanation: Compute the maximal allowed degrees in y for each unknown polynomial coefficient. These bounds are derived from the vanishing order and the geometry of the line at infinity. The formulas are:

$$dA = (m - 0 \cdot 2) \cdot \mathtt{d} - (m - 0 \cdot 3) \cdot 2 - 0 \cdot 3 + \mathsf{Twist}$$

$$dB = (m - 1 \cdot 2) \cdot \mathtt{d} - (m - 1 \cdot 3) \cdot 2 - 1 \cdot 3 + \mathsf{Twist}$$

$$dC = (m - 2 \cdot 2) \cdot \mathtt{d} - (m - 2 \cdot 3) \cdot 2 - 2 \cdot 3 + \mathsf{Twist}$$

$$dD = (m - 3 \cdot 2) \cdot \mathtt{d} - (m - 3 \cdot 3) \cdot 2 - 3 \cdot 3 + \mathsf{Twist}$$

Substituting $m = 7$, $\mathtt{d} = 16$, $\mathsf{Twist} = -7$:

- For $k = 0$: $dA = 7 \cdot 16 - 7 \cdot 2 - 0 - 7 = 112 - 14 - 7 = 91$
- For $k = 1$: $dB = 5 \cdot 16 - 4 \cdot 2 - 3 - 7 = 80 - 8 - 10 = 62$
- For $k = 2$: $dC = 3 \cdot 16 - 1 \cdot 2 - 6 - 7 = 48 - 2 - 13 = 33$
- For $k = 3$: $dD = 1 \cdot 16 - (-2) \cdot 2 - 9 - 7 = 16 + 4 - 16 = 4$

These bounds determine the maximum degree of the unknown polynomials in y.

```
dA := (m-0*2)*d - (m-0*3)*2 - 0*3 + twist:
dB := (m-1*2)*d - (m-1*3)*2 - 1*3 + twist:
dC := (m-2*2)*d - (m-2*3)*2 - 2*3 + twist:
dD := (m-3*2)*d - (m-3*3)*2 - 3*3 + twist:
```

Listing 4: Degree bounds

Part 5: Wronskian and Transition Formulas

Explanation: Define the Wronskian $\Delta_{yx} = y_1x_2 - y_2x_1$. The formula $Soly1$ expresses y_1 in terms of x_1, z_1, x_2, z_2 via the implicit function theorem applied to the hypersurface equation. It comes from differentiating the relation $RR(x, y, z) = 0$ with respect to the curve parameter and solving for y_1. The variables Rx, Ry, Rz are partial derivatives of RR that will be defined later.

```
Delta_yx := y1*x2 - y2*x1:
Soly1 := - x1*Rx/Ry - z1*Rz/Ry:
```

Listing 5: Wronskian and transition formulas

Part 6: Second Derivative Transition Formula

Explanation: Derive the expression for y_2 (denoted $Sol2$). This is obtained by differentiating the relation $Soly1$ with respect to the curve parameter and substituting the expressions for derivatives. The formula is lengthy but necessary for the jet transformation. It involves second partial derivatives of RR (Rxx, Rxy, Rxz, Ryy, Ryz, Rzz) that will be computed later.

```
Sol2 := - x1^2*Rxx/Ry + 2*x1^2*Rxy*Rx/Ry^2 + 2*x1*Rxy*z1*Rz/Ry^2 - 2*x1*z1*Rzz/Ry
        - Ryy*x1^2*Rx^2/Ry^3 - 2*Ryy*x1*Rx*z1*Rz/Ry^3 - Ryy*z1^2*Rz^2/Ry^3
        + 2*z1*Ryz*x1*Rx/Ry^2 + 2*z1^2*Ryz*Rz/Ry^2 - z1^2*Rzz/Ry
        - x2*Rx/Ry - z2*Rz/Ry:
```

Listing 6: Second derivative transition formula

Part 7: Transition to the Other Chart

Explanation: Substitute the expressions for y_1 and y_2 into J_{yx} to express it in terms of x_1, z_1, Δ_{zx} (the Wronskian in the alternative chart $\{R_y \neq 0\}$). The command `mtaylor` performs a multivariate Taylor expansion up to order 100, which is far beyond the needed degree. The result is sorted by the variables x_1, z_1, Δ_{zx}.

```
Jetzx := sort(mtaylor(Jetyx, [x1,z1,Deltazx], 100), [x1,z1,Deltazx]):
```

Listing 7: Transition to the other chart

Part 8: Extract Coefficients from Taylor Expansion

Explanation: Extract coefficients of specific monomials $x_1^{m-j-3k} z_1^j \Delta_{zx}^k$ from the Taylor expansion. For each triple $(i = m - j - 3k, j, k)$ we compute the coefficient. The factor $Rz^{m-j-3k} \cdot Ry^{m-2k}$ is multiplied to clear denominators, converting rational expressions into polynomial equations. The result is stored in $EEEq[i, j, k]$.

```
printflevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEEq[m-j-3*k,j,k] := factor(
            Rz^(m-j-3*k) * Ry^(m-2*k) *
            coeftayl(Jetzx, [x1,z1,Deltazx]=[0,0,0], [m-j-3*k,j,k])
        ):
    od:
od:
```

Listing 8: Coefficient extraction

Part 9: Define the Hypersurface Equation

Explanation: Define the specific hypersurface RR in $\mathbb{C}^3$ (affine chart). The actual surface degree is $\mathtt{d}+1=17$. The equation is:

$$RR = z^{17} + y^{17} + x^{17} + x^8 + 1 = 0$$

All first and second partial derivatives are computed symbolically for later use.

```
RR := z^(d+1) + y^(d+1) + TT;

Rx := diff(RR,x): Ry := diff(RR,y): Rz := diff(RR,z):

Rxx := diff(RR,x,x): Rxy := diff(RR,x,y): Rxz := diff(RR,x,z):
Ryy := diff(RR,y,y): Ryz := diff(RR,y,z): Rzz := diff(RR,z,z):
```

Listing 9: Hypersurface definition

Part 10: Expand Equations as Power Series in z

Explanation: Expand each $EEEq$ as a power series in z, truncating at order $(m-j-3k)\cdot\mathtt{d}$. This corresponds to the expected vanishing order along the hyperplane $\{z=0\}$. The truncation order is calculated as $(m-j-3k)\times 16$.

```
printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], z, (m-j-3*k)*d):
    od:
od:
```

Listing 10: Power series expansion in z

Part 11: Reduction Using the Surface Equation

Explanation: Create a basis for the ring of functions modulo the hypersurface equation. Using the relation $y^{\mathtt{d}+1} = -z^{\mathtt{d}+1} - TT$ (from $RR=0$), any power of y can be reduced to a polynomial in y of degree less than $\mathtt{d}+1$ with coefficients that are polynomials in x and z. The array $ny[j1]$ stores the reduced form of y^{j1}:

$$y^{j1} = (-z^{\mathtt{d}+1} - TT)^{\lfloor j1/(\mathtt{d}+1)\rfloor} \cdot y^{j1-(\mathtt{d}+1)\lfloor j1/(\mathtt{d}+1)\rfloor}$$

```
for j1 from 0 to 1+m*d+d do
    ny[j1] := (-z^(d+1)- TT)^(floor(j1/(d+1))) * y^(j1-(d+1)*floor(j1/(d+1))):
od:
```

Listing 11: Reduction of powers of y

Part 12: Define Unknown Coefficient Polynomials

Explanation: Initialize the unknown polynomials AA_j, BB_j, CC_j, DD_j as polynomials in y with undetermined coefficients. For each j (ranging from 0 to $m-3k$), the polynomial is expressed as a sum of terms $A||j[l]\cdot y^l$ where l runs from 0 to the degree bound. The symbols $A||j[l]$, etc., are unknown constants to be solved.

```
for l from 0 to m-0*3 do AA||l := add(A||l[j]*y^j, j=0..min(d, dA)); od:
for l from 0 to m-1*3 do BB||l := add(B||l[j]*y^j, j=0..min(d, dB)); od:
for l from 0 to m-2*3 do CC||l := add(C||l[j]*y^j, j=0..min(d, dC)); od:
for l from 0 to m-3*3 do DD||l := add(D||l[j]*y^j, j=0..min(d, dD)); od:
```

Listing 12: Initialize unknown polynomials

Part 13: Truncation in y

Explanation: Each equation is truncated as a power series in y up to degree $1+m\cdot\mathtt{d}+\mathtt{d} = 1+7\cdot 16+16 = 129$. This keeps computations manageable. The actual maximum degree needed is recorded in the array $dy[i,j,k]$.

```
printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], y, 1+m*d+d):
        dy[m-j-3*k,j,k] := degree(EEEq[m-j-3*k,j,k], y):
    od:
od:
```

Listing 13: Truncation in y

Part 14: Extract Coefficients in y

Explanation: For each equation, extract the coefficient of each power y^{j1} from the truncated polynomial. These coefficients, stored in $Cy[i,j,k,j1]$, are polynomial expressions in x and z (and the unknown constants) that must vanish identically.

```
printlevel := 3:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        for j1 from 0 to dy[m-j-3*k,j,k] do
            Cy[m-j-3*k,j,k,j1] := coeff(EEEq[m-j-3*k,j,k], y, j1):
        od:
    od:
od:
```

Listing 14: Extract y-coefficients

Part 15: Substitute Reduced y Powers

Explanation: Replace each power y^{j1} in the equations with its reduced form $ny[j1]$, which expresses it in terms of x, y, z modulo the hypersurface equation.

```
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEq[m-j-3*k,j,k] := add(Cy[m-j-3*k,j,k,j1]*ny[j1], j1=0..dy[m-j-3*k,j,k]):
    od:
od:
```

Listing 15: Substitute reduced powers of y

Part 16: Expand Unknown Polynomials in z (First Block)

Explanation: The coefficients $A||l[j]$ were originally constants. They are now expanded as polynomials in z. For the first block ($k=0$), each coefficient $A||l[j]$ is expressed as a sum over powers of z up to degree $dA-j$.

```
for l from 0 to m-0*3 do
    for j from 0 to min(d, dA) do
        A||l[j] := add(A||l[j,k]*z^k, k=0..dA-j):
    od:
od:
```

Listing 16: Expand coefficients in z for block A

Part 17: Expand Unknown Polynomials in z (Second Block)

Explanation: Expand the coefficients for the second block ($k = 1$) as polynomials in z.

```
for l from 0 to m-1*3 do
    for j from 0 to min(d, dB) do
        B||l[j] := add(B||l[j,k]*z^k, k=0..dB-j):
    od:
od:
```

Listing 17: Expand coefficients in z for block B

Part 18: Expand Unknown Polynomials in z (Third Block)

Explanation: Expand the coefficients for the third block ($k = 2$) as polynomials in z.

```
for l from 0 to m-2*3 do
    for j from 0 to min(d, dC) do
        C||l[j] := add(C||l[j,k]*z^k, k=0..dC-j):
    od:
od:
```

Listing 18: Expand coefficients in z for block C

Part 19: Expand Unknown Polynomials in z (Fourth Block)

Explanation: Expand the coefficients for the fourth block ($k = 3$) as polynomials in z.

```
for l from 0 to m-3*3 do
    for j from 0 to min(d, dD) do
        D||l[j] := add(D||l[j,k]*z^k, k=0..dD-j):
    od:
od:
```

Listing 19: Expand coefficients in z for block D

Part 20: Expand Equations as Power Series in z (After Substitution)

Explanation: After substituting the polynomial expansions in z, expand each equation again as a power series in z up to the required order.

```
printlevel := 2:
for k from 0 to m/3 do
    for j from 0 to m-3*k do
        EEq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):
    od:
od:
```

Listing 20: Second expansion in z

Part 21: Expand Unknown Polynomials in x (First Block)

Explanation: The coefficients $A||l[j,k]$ are expanded as polynomials in x. For the first block, each coefficient is expressed as a sum over powers of x up to degree $dA - j - k$.

```
printlevel := 3:
for l from 0 to m-0*3 do
    for j from 0 to min(d, dA) do
        for k from 0 to dA-j do
            A||l[j,k] := add(A||l[i,j,k]*x^i, i=0..dA-j-k):
        od:
    od:
od:
```

Listing 21: Expand coefficients in x for block A

Part 22: Expand Unknown Polynomials in x (Second Block)

Explanation: Expand the coefficients for the second block ($k = 1$) as polynomials in x.

```
printlevel := 3:
for l from 0 to m-1*3 do
    for j from 0 to min(d, dB) do
        for k from 0 to dB-j do
            B||l[j,k] := add(B||l[i,j,k]*x^i, i=0..dB-j-k):
        od:
    od:
od:
```

Listing 22: Expand coefficients in x for block B

Part 23: Expand Unknown Polynomials in x (Third Block)

Explanation: Expand the coefficients for the third block ($k = 2$) as polynomials in x.

```
printlevel := 3:
for l from 0 to m-2*3 do
    for j from 0 to min(d, dC) do
        for k from 0 to dC-j do
            C||l[j,k] := add(C||l[i,j,k]*x^i, i=0..dC-j-k):
        od:
    od:
od:
```

Listing 23: Expand coefficients in x for block C

Part 24: Expand Unknown Polynomials in x (Fourth Block)

Explanation: Expand the coefficients for the fourth block ($k = 3$) as polynomials in x.

```
printlevel := 3:
for l from 0 to m-3*3 do
    for j from 0 to min(d, dD) do
        for k from 0 to dD-j do
            D||l[j,k] := add(D||l[i,j,k]*x^i, i=0..dD-j-k):
        od:
    od:
od:
```

Listing 24: Expand coefficients in x for block D

Part 25: First Iteration — Divisibility by $z^{1\cdot \mathtt{d}}$

Explanation: In the first iteration, we require that the equations $Eq[1, m-1-3k, k]$ be divisible by $z^{1\cdot \mathtt{d}} = z^{16}$. We collect all coefficients (in x, y, z) from the Taylor expansion in z up to order $\mathtt{d} - 1 = 15$. These coefficients form a linear system in the unknown constants. The system is solved, and the solutions are assigned back to the unknowns.

```
Systemez1d := {}:
for k from 0 to (m-1)/3 do
    Coeffs_z1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,k], z, 1*d)), [x,y,z])}:
    Systemez1d := Systemez1d union Coeffs_z1d[1,m-1-3*k,k]:
od:
Systeme_z1d := Systemez1d:

Variables_z1d := indets(Systeme_z1d):
Solutions_z1d := solve(Systeme_z1d, Variables_z1d):
assign(Solutions_z1d):
```

Listing 25: First iteration: solving divisibility by $z^{1\cdot \mathtt{d}}$

Part 26: Output After First Iteration

Explanation: After solving, we output the number of terms remaining in each coefficient polynomial. This helps track the reduction process.

```
for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 26: Output after first iteration

Part 27: Second Iteration — Divisibility by $z^{2\cdot \mathtt{d}}$

Explanation: In the second iteration, we require divisibility by $z^{2\cdot \mathtt{d}} = z^{32}$. The equations considered are $Eq[2, m-2-3k, k]$. We expand in z up to order $2\mathtt{d} - 1 = 31$, collect coefficients, and solve the resulting linear system. The solutions from the previous iteration are already substituted via the `assign` command.

```
Systemez2d := {}:
for k from 0 to (m-2)/3 do
    Coeffs_z2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,k], z, 2*d)), [x,y,z])}:
    Systemez2d := Systemez2d union Coeffs_z2d[2,m-2-3*k,k]:
od:
Systeme_z2d := Systemez2d:

Variables_z2d := indets(Systeme_z2d):
Solutions_z2d := solve(Systeme_z2d, Variables_z2d):
assign(Solutions_z2d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 27: Second iteration: solving divisibility by $z^{2\cdot \mathtt{d}}$

Part 28: Third Iteration — Divisibility by $z^{3\cdot \mathtt{d}}$

Explanation: In the third iteration, we require divisibility by $z^{3\cdot \mathtt{d}} = z^{48}$. The equations considered are $Eq[3, m-3-3k, k]$. We expand in z up to order $3\mathtt{d} - 1 = 47$, collect coefficients, and solve.

```
Systemez3d := {}:
for k from 0 to (m-3)/3 do
    Coeffs_z3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,k], z, 3*d)), [x,y,z])}:
    Systemez3d := Systemez3d union Coeffs_z3d[3,m-3-3*k,k]:
od:
Systeme_z3d := Systemez3d:

Variables_z3d := indets(Systeme_z3d):
Solutions_z3d := solve(Systeme_z3d, Variables_z3d):
assign(Solutions_z3d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 28: Third iteration: solving divisibility by $z^{3\cdot \mathtt{d}}$

Part 29: Fourth Iteration — Divisibility by $z^{4\cdot \mathtt{d}}$

Explanation: In the fourth iteration, we require divisibility by $z^{4\cdot \mathtt{d}} = z^{64}$. The equations considered are $Eq[4, m-4-3k, k]$. We expand in z up to order $4\mathtt{d} - 1 = 63$, collect coefficients, and solve.

```
Systemez4d := {}:
for k from 0 to (m-4)/3 do
    Coeffs_z4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,k], z, 4*d)), [x,y,z])}:
    Systemez4d := Systemez4d union Coeffs_z4d[4,m-4-3*k,k]:
od:
Systeme_z4d := Systemez4d:

Variables_z4d := indets(Systeme_z4d):
Solutions_z4d := solve(Systeme_z4d, Variables_z4d):
assign(Solutions_z4d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 29: Fourth iteration: solving divisibility by $z^{4\cdot \mathtt{d}}$

Part 30: Fifth Iteration — Divisibility by $z^{5\cdot \mathtt{d}}$

Explanation: In the fifth iteration, we require divisibility by $z^{5\cdot \mathtt{d}} = z^{80}$. The equations considered are $Eq[5, m-5-3k, k]$. We expand in z up to order $5\mathtt{d} - 1 = 79$, collect coefficients, and solve.

```
Systemez5d := {}:
for k from 0 to (m-5)/3 do
    Coeffs_z5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,k], z, 5*d)), [x,y,z])}:
    Systemez5d := Systemez5d union Coeffs_z5d[5,m-5-3*k,k]:
od:
Systeme_z5d := Systemez5d:

Variables_z5d := indets(Systeme_z5d):
Solutions_z5d := solve(Systeme_z5d, Variables_z5d):
assign(Solutions_z5d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 30: Fifth iteration: solving divisibility by $z^{5\cdot \mathtt{d}}$

Part 31: Sixth Iteration — Divisibility by $z^{6\cdot \mathtt{d}}$

Explanation: In the sixth iteration, we require divisibility by $z^{6\cdot \mathtt{d}} = z^{96}$. The equations considered are $Eq[6, m-6-3k, k]$. We expand in z up to order $6\mathtt{d} - 1 = 95$, collect coefficients, and solve.

```
Systemez6d := {}:
for k from 0 to (m-6)/3 do
    Coeffs_z6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,k], z, 6*d)), [x,y,z])}:
    Systemez6d := Systemez6d union Coeffs_z6d[6,m-6-3*k,k]:
od:
Systeme_z6d := Systemez6d:

Variables_z6d := indets(Systeme_z6d):
Solutions_z6d := solve(Systeme_z6d, Variables_z6d):
assign(Solutions_z6d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 31: Sixth iteration: solving divisibility by $z^{6\cdot \mathtt{d}}$

Part 32: Seventh Iteration — Divisibility by $z^{7\cdot \mathtt{d}}$

Explanation: In the seventh iteration, we require divisibility by $z^{7\cdot \mathtt{d}} = z^{112}$. The equations considered are $Eq[7, m-7-3k, k]$. We expand in z up to order $7\mathtt{d} - 1 = 111$, collect coefficients, and solve. After this iteration, the output shows that each coefficient polynomial has been reduced to a single constant term (all `nops` outputs are 1).

```
Systemez7d := {}:
for k from 0 to (m-7)/3 do
    Coeffs_z7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,k], z, 7*d)), [x,y,z])}:
    Systemez7d := Systemez7d union Coeffs_z7d[7,m-7-3*k,k]:
od:
Systeme_z7d := Systemez7d:

Variables_z7d := indets(Systeme_z7d):
Solutions_z7d := solve(Systeme_z7d, Variables_z7d):
assign(Solutions_z7d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 32: Seventh iteration: solving divisibility by $z^{7\cdot \mathtt{d}}$

Part 33: Eighth Iteration — Divisibility by $z^{8\cdot \mathtt{d}}$

Explanation: The code continues to higher orders. The eighth iteration requires divisibility by $z^{8\cdot \mathtt{d}} = z^{128}$. The pattern is identical to previous iterations.

```
Systemez8d := {}:
for k from 0 to (m-8)/3 do
    Coeffs_z8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,k], z, 8*d)), [x,y,z])}:
    Systemez8d := Systemez8d union Coeffs_z8d[8,m-8-3*k,k]:
od:
Systeme_z8d := Systemez8d:

Variables_z8d := indets(Systeme_z8d):
Solutions_z8d := solve(Systeme_z8d, Variables_z8d):
```

```
assign(Solutions_z8d):
```

Listing 33: Eighth iteration: solving divisibility by $z^{8\cdot \mathtt{d}}$

Part 34: Ninth Iteration — Divisibility by $z^{9\cdot \mathtt{d}}$

Explanation: The ninth iteration requires divisibility by $z^{9\cdot \mathtt{d}} = z^{144}$.

```
Systemez9d := {}:
for k from 0 to (m-9)/3 do
    Coeffs_z9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,k], z, 9*d)), [x,y,z])}:
    Systemez9d := Systemez9d union Coeffs_z9d[9,m-9-3*k,k]:
od:
Systeme_z9d := Systemez9d:

Variables_z9d := indets(Systeme_z9d):
Solutions_z9d := solve(Systeme_z9d, Variables_z9d):
assign(Solutions_z9d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 34: Ninth iteration: solving divisibility by $z^{9\cdot \mathtt{d}}$

Part 35: Tenth Iteration — Divisibility by $z^{10\cdot \mathtt{d}}$

Explanation: The tenth iteration requires divisibility by $z^{10\cdot \mathtt{d}} = z^{160}$.

```
Systemez10d := {}:
for k from 0 to (m-10)/3 do
    Coeffs_z10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10-3*k,k], z, 10*d)),
        [x,y,z])}:
    Systemez10d := Systemez10d union Coeffs_z10d[10,m-10-3*k,k]:
od:
Systeme_z10d := Systemez10d:

Variables_z10d := indets(Systeme_z10d):
Solutions_z10d := solve(Systeme_z10d, Variables_z10d):
assign(Solutions_z10d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 35: Tenth iteration: solving divisibility by $z^{10\cdot \mathtt{d}}$

Part 36: Eleventh Iteration — Divisibility by $z^{11\cdot \mathtt{d}}$

Explanation: The eleventh iteration requires divisibility by $z^{11\cdot \mathtt{d}} = z^{176}$. This is the last iteration that yields a nontrivial solution.

```
Systemez11d := {}:
for k from 0 to (m-11)/3 do
    Coeffs_z11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11-3*k,k], z, 11*d)),
        [x,y,z])}:
    Systemez11d := Systemez11d union Coeffs_z11d[11,m-11-3*k,k]:
od:
Systeme_z11d := Systemez11d:
```

```
Variables_z11d := indets(Systeme_z11d):
Solutions_z11d := solve(Systeme_z11d, Variables_z11d):
assign(Solutions_z11d):

for I from 0 to m - 0*3 do nops(expand(AA||I)) od;
for I from 0 to m - 1*3 do nops(expand(BB||I)) od;
for I from 0 to m - 2*3 do nops(expand(CC||I)) od;
for I from 0 to m - 3*3 do nops(expand(DD||I)) od;
```

Listing 36: Eleventh iteration: solving divisibility by $z^{11\cdot \mathrm{d}}$

Part 37: Final Verification

Explanation: After completing all iterations, we reconstruct the final expression for the jet differential J_{yx} using the solved coefficients. We then compute the set of all remaining variables (excluding the coordinate variables $x, x1, x2, y, y1, y2, z$). If the output is 0, this means there are no free parameters remaining, confirming that all polynomial coefficients have been determined to be zero, and consequently the only solution is the trivial one. Hence, no nonzero negatively twisted invariant jet differential exists for these parameters.

```
Jetyx := factor(
    + (1/Rz^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (y1)^j * Delta_yx^0, j=0..m-0*3)
    + (1/Rz^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (y1)^j * Delta_yx^1, j=0..m-1*3)
    + (1/Rz^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (y1)^j * Delta_yx^2, j=0..m-2*3)
    + (1/Rz^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (y1)^j * Delta_yx^3, j=0..m-3*3)
):

Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:
NbVars := nops(Vars);
```

Listing 37: Final verification

Conclusion

The Maple code systematically constructs an invariant jet differential of order 7 on a degree 17 hypersurface in $\mathbb{P}^3$ with a twist of -7. The solution is obtained by imposing vanishing conditions order by order in z and solving linear algebraic systems. The final verification shows that all undetermined coefficients are forced to be zero, demonstrating that no nonzero such differential exists for these parameters.